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SafeCity — A GIS-based tool profiled for supporting decision making in urban development and infrastructure protection

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ABSTRACT

This paper presents a system for analysis of municipal Critical Infrastructures, which offers integrated tools for target analysis, hazard scenario simulations and spatial analysis within a remotely accessible Web-based Geographic Information System. The system has been applied to research conducted in the city of Gdansk with the aid of blast attack, chemical leakage and flood hazard scenarios, as well as a spatial density algorithm, which highlights events in which the proximity of infrastructures influences their susceptibility to a single attack. The paper also discusses the way in which the tools provided by the system aim to assist in the processes of infrastructure vulnerability assessment, mitigating discovered risks as well as strategic planning of city development.

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1. Introduction

The importance of infrastructures in the proper functioning of a city cannot be overestimated. Ever since the dawn of the industrial era, a well developed infrastructure decided the success or failure of a city's economic growth [1]. The natural consequence of infrastructure development was the creation of local clusters of interconnected services and companies [2]. This local concentration of different infrastructure sectors did not come into wider attention until the second half of the XX century, when the aftermath of World War II saw a decentralization of urban areas in the United Kingdom [3]. However, this was mainly a local trend. Globally the rapid evolution of telecommunication networks has led to the creation of a new type of infrastructure which controls access to all sorts of goods and services via software mechanisms such as passwords or biometric judgments [4]. The internet itself has been designed, in accordance with the post-war paradigms, as a dispersed network able to withstand a direct physical attack [5]. However, while the entire network may be resistant to most types of hazards, a local malfunction of several nodes (caused e.g. by a power shortage) may be enough to disconnect a small city, the consequences of which may be quite severe [6]. By the end of the last century the impact of technological development on modern municipalities has become so great that certain types of infrastructure have been recognized as essential to the proper functioning of a city.

The first formal definition of Critical Infrastructure was produced by the United States government in Executive Order 13010 issued in 1996. The document defines Critical Infrastructures as "independent, mostly privately-owned, man-made systems that function collaboratively and synergistically to produce and distribute a continuous flow of essential goods and services". The provided services must be vital enough so that "incapacity or destruction" of one of the infrastructures would have a "debilitating impact" on citizen security [7]. This definition was later expanded in the USA National Strategy on Homeland Security (NSHS), published in 2002. This document not only contains a list of Critical Infrastructure sectors, but also makes a distinction between Critical Infrastructures and key assets, which are defined as individual targets whose "destruction would not endanger vital systems, but could create local disaster or profoundly damage" the "morale and confidence" of citizens. Such assets would include historical attractions (national, state, and local

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monuments and icons) and other localized facilities with destructive potential or of high value to a community such as schools, courthouses and bridges [8]. Three years later, in Europe the Green Paper on the European Programme for Critical Infrastructure Protection (EPCIP) provided the definitions of Critical Infrastructures on different levels (e.g. National and European), the definition of their threats (a "threat" being any person, activity, or event with the potential to cause harm to a system or operational environment — the primary types of threats include natural disasters, accidental threats and threats of intentional or malicious nature [9]) as well as the role of their owners and operators. Both these documents also contain lists of Critical Infrastructure sectors. These, in general, are largely similar and consist of Energy, Water, Food, Public Health, Financial, Transport, and Chemical industry, Telecommunications and Research facilities [10].

The successful protection of Critical Infrastructure is a complex process, encompassing the events occurring before a threat is identified, the time during a crisis and its aftermath. Consequently, the main phases of Critical Infrastructure emergency management are referred to as Preparedness, Mitigation, Response and Recovery. As the naming indicates, the first phase involves threat identification, assessment of risk and planning of response actions, as well as training of emergency services [11]. The next phase concentrates on attempts to prevent hazards from developing into disasters, as well as reducing the effects of disasters when they occur. Mitigatory measures may involve structural means such as perimeter fencing or flood dykes, or non-structural measures such as legislation (e.g. regulating evacuation or methods of alarming the public), land-use planning (e.g. the designation of non-essential land to be used as flood zones) and insurance [12]. The Response phase takes place immediately after a disaster strikes, and includes the mobilization of necessary emergency services and first responders in the affected area. A well rehearsed emergency plan developed as part of the preparedness phase is paramount to the efficiency of the search and rescue efforts, as the fatality rates increase dramatically after the initial 72 h after impact [13]. The efforts of the final phase aim to restore the affected area to its original state, and involve rebuilding destroyed assets and repair of other essential infrastructures [14].

In the last decade the issue of Critical Infrastructure protection has received a lot of attention from researchers worldwide. Soon after the events of 9/11, it was proposed that the security of Critical Infrastructure could be improved by development of existing supervisory control and data acquisition systems (SCADA) [15]. However, it soon transpired that the rising protection costs stemming from the quick evolution of existing as well as new threats, e.g. related to cyber security, make it impossible to completely secure any given infrastructure [16].

Later research suggested that Critical Infrastructures should not be analyzed individually, but rather as a network of distributed interdependent systems [17]. The tight interconnections between these systems could, in a worst-case scenario, lead to a cascading failure [18]. Evidence supporting this theory could be found in the wake of crises such as Hurricane Katrina [19]. In consequence, new models and methods of Critical Infrastructure risk analysis have been proposed, including probabilistic models [20], graph models [21], economical models [22,23], agent-based models [24] and network models [25], as well as empirical approaches [26].

A common characteristic of all the aforementioned methodologies is that they have been developed for use by professional analysts. The task of constructing an analysis and presenting results in a meaningful way rests on the user. The available tools also relatively seldom [21,23,25] consider the geographical location of infrastructures as a factor affecting their security.

This study aims to prove that modern technology in the form of a well-proven vulnerability assessment methodology coupled with a user-friendly Geographic Information System (GIS) and tools for Geovisual Analytics may help in significantly enhancing the security of Critical Infrastructures in medium as well as small cities which need to cope with a very limited budget for infrastructure protection.

The responsibility for protecting Critical Infrastructures may be subject to local legislation, but most often it lies on the owner and/or operator. Most often at least one of these roles is attributed to a local municipality. However, municipalities often lack the expertise and resources necessary to ascertain appropriate level of protection to the right infrastructures, many of which are used in ways that were not foreseen during their development [27]. Thus, successful management of risk for a large number of different types of Critical Infrastructure with variable resistance to various types of threats requires support from dedicated tools employing specialistic knowledge from many branches in support of integrated analysis and comparison of different types of municipal infrastructure.

2. Critical Infrastructure analysis and risk assessment

The first step in protecting Critical Infrastructure involves identifying and evaluating the factors that may negatively influence its operations. This process is referred to as risk assessment [28] or Security Vulnerability Assessment (SVA) [29]. Although there are numerous methods of assessing risk, they all share a common set of variables. These characteristics are defined as follows [30]:

Vulnerability v_i is the likelihood that an inherent weakness in a system or its operating environment may be exploited to cause harm to the system. It is expressed as a probability p_i that component i will fail [31]: $v_i = p_i$. Vulnerability is often expressed in percentage and thus ranges from 0% to 100%.

Damage d_i is the cost of a fault, expressed in terms of casualties, loss of productivity, loss of capital equipment, loss of time and so forth. It is typically expressed in currency.

Risk r_i denotes the possibility of incurring a certain amount of damage. Therefore, the risk for component i equals: $r_i = p_i * d_i$. Like damage, risk may also be expressed in terms of monetary loss.

Availability a_i is the probability that a component is operational at any point in time. Therefore $a_i = (1 - p_i)$, which means that availability is the complement of the probability of a fault.

Criticality represents the combined value of a component, measured in the number of lives endangered by an attack directed against it as well as the damage

caused by its incapacitation combined with the costs of restoring it to full functionality.

The American Society of Mechanical Engineers (ASME) defined several equations for determining the levels of risk [32]. The most commonly applied formula:

$$R_{ai} = F_{ai}(\text{Vulnerability})_{ij}(\text{Consequences})_{ij} \quad (1)$$

involves the use of the following variables:

R_{ai} denotes the annual economic risk for a given threat i;

F_{ai} represents the annual frequency of an adversary attacking a critical asset using specific type of threat i; If F_{ai} equals "1.0", then the calculated risk is referred to as "conditional threat risk";

$(\text{Vulnerability})_{ij}$ is the conditional probability that a specific failure mode j will occur, assuming the occurrence of potential threat i;

$(\text{Consequences})_{ij}$ equals the total measure of consequences of failure for threat i failing in mode j.

There are various models and methodologies for assessing risk. However, basing on the extent of analysis, the required resources (which can vary depending on the scope of the assessment), the availability of reliable data on risk factors as well as the involved methods of analysis, they may be divided into two general categories referred to as qualitative and quantitative methods. Quantitative approaches (generally) estimate a monetary value associated with a risk and its reduction based on factors such as the likelihood that a damaging event will happen, the costs resulting from potential losses from the event and the costs necessary for mitigating actions resulting from those losses. The cost of implementing countermeasures is then compared to the cost of replacing lost assets and in order to determine the cost-effectiveness of the countermeasure. In an event where reliable data and costs cannot be determined, an analyst may employ qualitative measures, which define risk in subjective terms of "high," "medium," and "low" depending on their potential impact on the infrastructure's operation. However, qualitative assessments are generally discouraged because they rely upon the expertise, experience, and judgment of the analysts, and thus may be subject to skewed perspectives of the party responsible for conducting the assessment [33].

3. The CARVER2™ software for Critical Infrastructure risk assessment

Over the years many different organizations made their attempts at creating a software which would in various degrees assist analysts in the process of conducting risk assessment. An international survey on Critical Infrastructure interdependency modeling, performed by the U.S. Department of Energy National Laboratory in August 2006, lists thirty software packages aimed at supporting different aspects of infrastructure vulnerability assessment, including modeling and simulation of their interactions and interdependencies [34]. Several applications from that list were considered for use during the development of the presented system.

In order to fulfill the goals of the study, the Critical Infrastructure analysis module of the system had to be based on proven methodology and needed to provide support for major infrastructure sectors in the city of Gdansk, which include energy sources and chemical storage. The tool needed to be platform-independent and affordable, preferably based on Open Source technology for easy integration with other system modules. Because Gdansk does not employ trained experts in Critical Infrastructure protection, it was desirable that the tool has an easy to use Graphical User Interface (GUI).

This established set of rules dictated that several well-proven systems using Open Source technologies such as the Electrical Distribution Design [35] or TRANSIMS [36] could not be utilized because of being dedicated to the analysis of a certain Critical Infrastructure sector.

More universal solutions offered by Agent-Based infrastructure modeling systems based on open technologies were also considered [37,38], however, at the time of the system's development neither left prototype phase.

A universal multi-platform analysis tool called Critical Infrastructure Modeling Simulation (CIMS) [39] was under consideration as well. This software package allows the user to visually analyze the interoperability of several infrastructure sectors. The application supports rapid infrastructure model development basing on open source data. Unfortunately, the greatest advantages of CIMS also constitute its biggest drawback, which is the necessity of manual construction of the Critical Infrastructure network model.

Finally, the Inoperability Input–output Model (IIM) was considered. IIM is a mature analytical framework tuned for assessing the impacts of infrastructure economical interdependencies [40]. Unfortunately, the IIM was developed with focus on the United States economy and uses data from the U.S. Department of Commerce for estimating the impacts of Critical Infrastructure disruptions. While adapting the model for more general use would certainly be possible, it would be far out of the scope of the presented work.

Ultimately, the CARVER package was selected because it supports analysis of many infrastructure sectors, through a user-friendly GUI, and using Open Source technologies. This meant that it delivered the necessary functionalities and could be easily integrated with other elements of the system. It is also based on a proven methodology in the form of the C.A.R.V.E.R. Matrix.

The C.A.R.V.E.R. Matrix is a decision tool developed by U.S. Military for rating the relative attractiveness of potential targets and for properly allocating attack resources. The C.A.R.V.E.R. risk assessment algorithm analyses and evaluates factors such as Criticality, Accessibility, Recuperability, Vulnerability, Effect and Recognizability for every potential target. Each of the analyzed targets forms part of a decision matrix, which indicates which one of them achieves the highest ranking within the given requirements [41].

The procedure by which an attacking force chooses the proper object for attack is referred to as target analysis. In general, target analysis allows attackers to identify the most effective type of attack which may be successfully carried out with least possible effort [42].

With CARVER2™, the National Infrastructure Institute (NI^2) applied target analysis to identification of weaknesses in the defense of Critical Infrastructures and their surroundings. This way the software attempts to "reverse engineer" an attack scenario in order to devise the most effective countermeasures. By employing quantitative methods of analysis to a set of standard attributes, CARVER2™ allows for comparing dissimilar types of Critical Infrastructure using the same standards. It is designed for quick and easy identification and comparison of potential terrorist targets at the local, state and national levels in order to assist government officials in the allocation of protective resources.

In the process of adapting the C.A.R.V.E.R. Matrix to civilian use, NI^2 modified the original assessment factors [43]. Thus, the CARVER2™ assessment algorithm involves a considerable number of attributes evaluated in several categories. Factors such as the number of people affected from the loss in a 72 hour period, the cost to rebuild or replace the structure and associated potential deaths are analyzed within the scope of Criticality. The relative ease by which an attacking force may enter or approach the Critical Infrastructure is evaluated within the Accessibility category. The amount of time required to repair or replace the Critical Infrastructure (the longer the time needed to replace the infrastructure, the more attractive target it poses to an attacking force) is defined within the Recoverability category. Factors such as the ease at which various infrastructure types might be destroyed and their inhabitants killed by different types of attack (blast or chemical by default) are evaluated in the scope of Vulnerability. The relative notoriety of a facility (to what degree the facility is a key asset) is reflected by the Espyability category. The relative percentage of the facility's functions that can be backed up is described within the Redundancy category. The CARVER2™ algorithm also attempts to identify the type of the examined infrastructure and represent its influence on infrastructures in different sectors. These factors are analyzed within a separate category, referred to as Interdependency.

For every investigated Critical Infrastructure the CARVER2™ calculates a synthetic score which reflects the total value of all data inputs and gives a numerical basis for making informed comparisons between different infrastructures. This unified approach allows for comparing the relative vulnerability of even drastically different infrastructure components, which makes the algorithm suitable, for example, for supporting decisions on the optimum allocation of limited resources for best effect on overall security.

Thus, the CARVER2™ package provides a simple, non-technical PC-based tool that analyses and ranks critical infrastructures in order of importance, offers basis for making appropriate decisions on protecting them and allows for "cross-sector" scoring and ranking, while taking into account the strategic objectives of the Homeland Security directive and retaining the well-proven foundation of recognized methodology used by Secret Service and U.S. military [44].

4. The role of Geographical Information Systems and Geovisual Analytics in hazard management

Because Critical Infrastructures are an element of urban areas, a significant part of the information required to perform the process of their risk analysis is located in geospatial databases held by the municipal offices. These databases are created and maintained with the help of dedicated software, called a Geographic Information System. For this reason, it would seem logical to use GIS as an aid in the process of vulnerability analysis. Unfortunately, for various reasons, the application of Geographic Information Systems in the hazard management process is limited to simple mapping and monitoring of detected threats, sometimes accompanied by simulation of their expected progress [45]. Numerical data sets produced by tools supporting the process of risk analysis do not posses their own geographical context, which prevents their easy integration with Geographic Information Systems. Moreover, current Critical Infrastructure analysis software such as CARVER2™ does not take into account the spatial interdependencies between investigated infrastructures. At the same time the assessment of Critical Infrastructure vulnerability based strictly on tabular data is counter-intuitive and often incomplete. It is also time-consuming because it requires an interdisciplinary collaboration between persons involved in various aspects of security.

The field of science which deals with tasks requiring time-sensitive collaboration, analytical reasoning and dissemination of information to a variety of audiences is called Visual Analytics. According to Thomas and Cook, the concept of Visual Analytics may be applied to construct software tools which would support sense making, cognition, and perception [46]. In 2007 Andrienko et al. proposed the application of these guidelines to the development, evaluation and selection of proper policies, plans, scenarios, projects or interventions where the problems have a geographic or spatial component. This concept, which involves the support of human abilities of vision and cognition with computer-based visual interfaces that provide flexible connections to relevant data and are specifically designed to provide support for analytical reasoning, was named Geovisual Analytics [47].

Because Geovisual Analytics can be used to support geospatial intelligence [48], situation awareness [49], and other tasks that require analysis and knowledge construction using geospatial information, Tomaszewski et al. proposed its application to Crisis Management [50]. He supports this concept by stating that Geovisual Analytics can aid collaboration processes where dynamic, time critical response situations require actors to engage in tasks for which they may not have procedural training or experience or where information is fragmentary or incomplete. Although the concept of implementing Geovisual Analytics to crisis management seems to be logically solid, it has not been actually implemented in a working system thus far, and therefore its efficiency in application to crisis management has not been verified. In addition, the current state-of-the-art does not deal with risk assessment and mostly neglects simulation of hazard scenarios.

Summing up the current state-of-the-art, the applications that have been thus far used to support risk assessment do not take into account the possibility of geographical representation and spatial analysis of the results. At the same time, the recently developed tools dedicated to geospatial visualization do not operate on Critical Infrastructure vulnerability data and most often do not allow for simulation of threat scenarios.

In this context, the presented system aims to converge risk assessment software with GIS technology in order to provide an integrated tool which would enable management,

processing and spatial analysis of infrastructure vulnerability as well as investigation of its geographical interactions and dissemination of the results in an accessible form to the decision makers.

5. The system architecture

The SafeCity Geographic Information System for Critical Infrastructure protection builds upon the Web-based Geographic Information System for Assessment and Visualisation of Critical Infrastructure and its Hazards by Civilian Services, originally developed by the Gdansk University of Technology (GUT) in the course of the European Programme for Critical Infrastructure Protection. The system enables operators to identify Critical Infrastructures in the context of multi-layer maps of a city, perform target analysis and simulate different hazard scenarios. Once every identified Critical Infrastructure has been assigned a criticality rank, the system allows to perform spatial analysis of the density of vulnerable structures across different areas of the city.

The system is built around a client–server architecture, with different functionalities delegated to separate, semi-independent modules, as shown in Fig. 1.

The City Database contains non-spatial attributes of geographical data, such as the type and role of a building, as well as its number of inhabitants. It is not considered an integral part of the system, but instead an optional external source of data. It represents the system's interface for dealing with existing databases (possessed by virtually all municipalities), which may not be fit for direct interaction with GIS, but nevertheless contain valuable information which could benefit the process of vulnerability analysis.

The Web-based Access Portal serves as a gateway to the system. Built with Java Server Pages as well as basic DHTML technology, it performs user authentication and generates most standard elements of the Graphical User Interface.

The CARVER2™ module implements target analysis algorithms based on C.A.R.V.E.R. methodology in order to allow vulnerability analysis of selected structures, using information provided by experts as well as available non-spatial attributes (such as the level of security of a building) extracted from city databases. It produces synthetic scores regarding the vulnerability of Critical Infrastructures which are attributed to features representing the investigated structures in the Geodatabase.

The Geodatabase is the central storage module of the system. It stores information regarding infrastructure vulnerability as well as results of hazard scenario simulations and spatial analyses. It serves as a source of data for both the latter, and the geographic data stored within it is processed into thematic layers by the GeoServer.

The hazard scenario simulation module enables the execution of threat models in the context of available geospatial data. Thus far the implemented hazard models encompass blast attack, chemical leakage and flooding. The models use information such as terrain elevation, building layout and (point or line-based) source of threat in order to generate a thematic layer representing the outcome of a given disaster.

The Spatial Analysis Engine is designed to perform a multitude of operations which support Geovisual Analytics of Critical Infrastructure vulnerability. This involves interpolation of CARVER2™ criticality ranks and generating maps of risk for different types of attack. The generated analyses are stored within the geodatabase so that they may be accessed and disseminated through the GeoServer.

The GeoServer is the central point of the system, from which all geographical data is served to the end users. The module itself is a direct adaptation of a dynamically developing Open Source GIS server with full support for Open Geospatial Consortium's (OGC) open data exchange protocols such as Web Map Service (WMS), Web Coverage Service (WCS), Web Processing Service (WPS) and Web Feature Service Transactional (WFS-T) [51]. The GeoServer also provides real-time data reprojection, which allows e.g. to overlay data created in

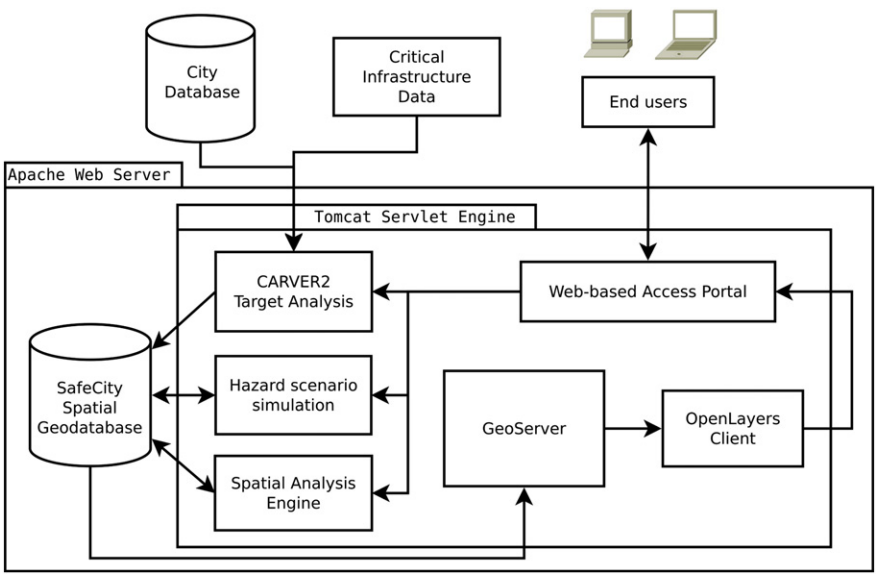

Fig. 1. Overview of *SafeCity* architecture.

local coordinate systems on satellite images provided by services such as Google Maps [52].

The GIS client, which runs from within a web browser, is based on a JavaScript library for displaying map data called OpenLayers. OpenLayers implements a constantly developing JavaScript Application Programming Interface (API) for building rich web-based geographic applications with no server-side dependencies. Although it behaves similarly to API's such as Google Maps and MS Bing Maps, OpenLayers is Open Source, which makes it free for modification and redistribution. In addition, it supports open protocols such as OGC WMS and WFS, which makes it an excellent companion to the GeoServer [53]. The client provides basic functionality such as navigating the map and switching visible layers, while also acting as a starting point for launching hazard simulations and spatial analyses.

It is worth noting that the system was developed entirely with the use of Open-Source technology. In addition to GeoServer and OpenLayers, the system incorporates the Apache Web Server, the Tomcat Servlet Engine and custom Java Platform code. This makes the system both cost-efficient as well as fully adaptable to any configuration of hardware and software which may be used by interested municipalities.

6. Applications of SafeCity system to the analysis of Critical Infrastructures in the city of Gdansk

This section discusses the results of several different analyses conducted with the aid of SafeCity in the area of the city of Gdansk. With an area of 262 km^2 and nearly half a million residents, Gdansk is the largest city in northern Poland. Over a thousand years of history make the city's infrastructure well developed, however the aftermath of World War II has left large areas of the city in need of new land use plans. These qualities make the city a very good place to exemplify the system's tools for both analysis of the existing Critical Infrastructure as well as investigation of planned locations for new structures.

6.1. Analysis of infrastructure vulnerability to a single blast attack

Thermal explosions are complex processes which involve chemical and physical reactions such as heating, expansion, and phase transition as well as transfer of heat and momentum. The main destructive force of an explosive blast is high overpressure produced by the rapidly expanding wave front of the explosion. Depending on the strength of the explosion, the wave front may knock over, severely damage or even shatter all objects within a given proximity to the center of explosion [54]. Because of this destructive potential, high explosives have received a lot of interest from the military ever since their introduction at the end of the 19th century. The conducted experiments suggested the possibility to relate the explosive patterns observed in many major explosion incidents to equivalent TNT-charge weights by converting the combustion energy in the investigated material into an equivalent charge weight of TNT. The factor used to relate this material energy to TNT is commonly referred to as the equivalence factor [55]. Basing on field test results, mathematical models of the released energy have been created for many different materials.

The blast model used by the SafeCity hazard simulation module is derived from the algorithm for TNT-based blast simulation obtained from the tests performed by the U.S. Department of the Army [56]. Using the data sent by the client, which includes the point of placement as well as type and mass of explosive material, the module generates vector layers containing the simulation results.

The following analysis depicts a simulated terrorist attack on the City Hall, involving a small van containing 500 kg of Ammonium Nitrate Fuel Oil (ANFO). The van is parked in the small area outside of the main building, as shown in Fig. 2.

When appropriate data regarding the placed charges has been input, the Web GIS client sends it to the blast simulation module. The module generates a vector layer representing areas with different exposures to the pressure generated by the explosion. Complying with the paradigms of Geovisual Analytics, the areas within the layer are colored according to the strength of simulated overpressure. In the case of the presented analysis, as shown in Fig. 3, the area within 12 m from the center of the explosion, marked in red, represents overpressure of over 20 kPa which incurs structural damage and certain fatalities. The next region, denoted in orange, covers an area within 61 m from the center of explosion. In this area people and structures would be exposed to peak overpressure greater than 13 kPa, which could result in moderate damage to structures as well as potentially lethal injuries. The largest region, colored in yellow, represents the area where flying glass shards may incur minor damage to life and property. The place of minimum stand-off, which would allow one to avoid the direct consequences of the blast (corresponding peak overpressure below 6 kPa), would be about 300 m from its center [57].

Because the simulation results are created in the form of GIS layers, the system allows for their remote geospatial analysis. By default, the Spatial Analysis module of the system detects when a spatial threat simulation is started and automatically performs an associated analysis. The results are immediately available to the end user in the form of thematic layers associated with the particular simulation. In the case of the presented analysis, the module automatically marked the structures which fall in range of the simulated blast, issued a query to the city database, and returned a new layer containing the affected structures along with their details and estimated number of residents. The results are shown in the window in the right part of Fig. 3.

The presented simulation clearly shows that such an attack would not only devastate the City Hall, but also severely damage both the court building as well as the main police station, which are located on the opposite side of the street. Moreover, the analysis involving non-spatial data from these institutions reveals that the total number of potential victims of such an attack could be counted in hundreds, as the estimated number of residents in the buildings closest to the center of the explosion adds up to 680 people. Although not all of them may be there when the attack takes place, in reality the number could be even higher, as the simulation does not take into account the possible number of people on the street (which is one of the busiest in the city).

The conducted analysis shows that the architectural layout of the city, which conveniently places several important institutions close to one another near a transportation hub,

Fig. 2. Placement of 500 kg charge outside of the City Hall via the *SafeCity* administration panel.

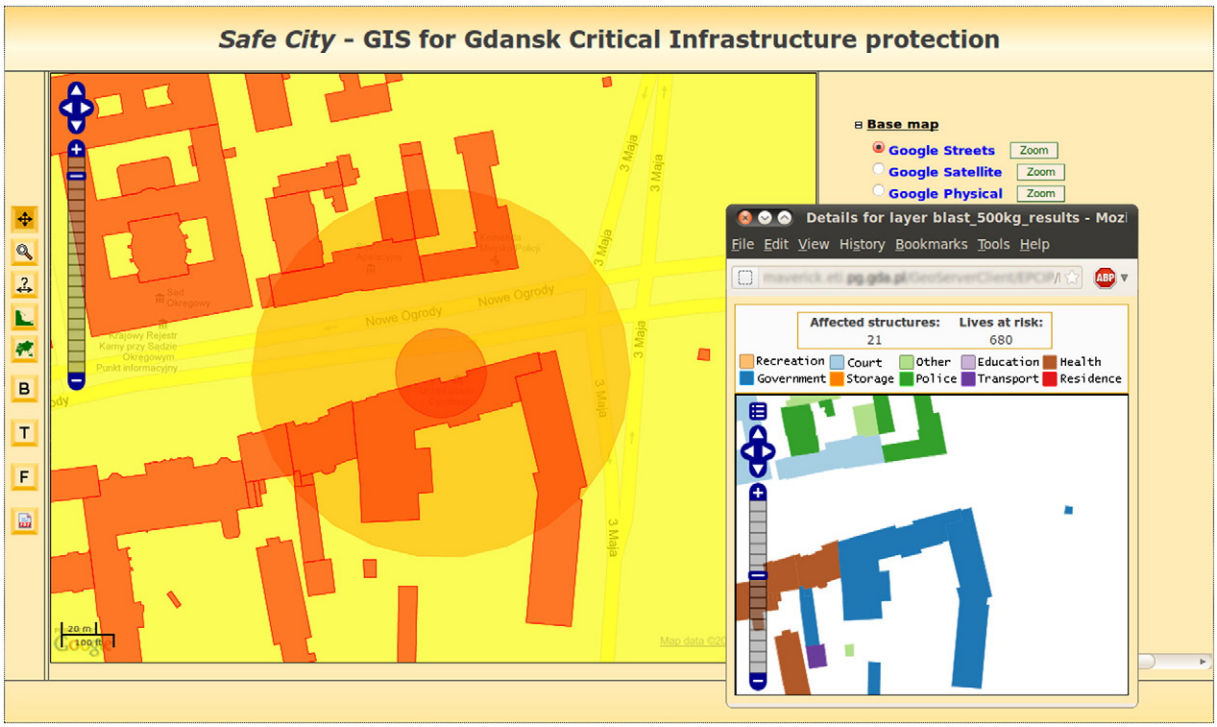

Fig. 3. Results of the blast simulation with a detailed report on affected structures shown in a separate window.

constitutes a serious vulnerability issue in the event of a terrorist attack. Such geographical clustering makes all of these structures susceptible to a single blast attack which could adversely affect their continued operation.

6.2. Analysis of the effects of a chemical vapor cloud on the neighborhood of a chemical plant

The behavior of a cloud of chemicals over a given terrain and under specific weather conditions may be forecast with the use of dedicated atmospheric dispersion models. These models analyze the spread of volatile substances from their point of release to the final downwind concentration of interest. In principle, for a typical pressurized fluid, this process consists of an initial momentum-dominated phase, followed by a dense gas phase determined by gravitational effects, and concludes with passive neutral buoyancy affected primarily by atmospheric turbulence. Dispersion models may also specialize in simulating a certain type of release. These are often categorized as continuous or instantaneous, however in some cases instantaneous spills may involve a large initial puff followed by a continuous release generated from an evaporating pool [58].

One of the most popular and well-proven models for describing far-field, long time-scale behavior of steady, continuous emissions is the Gaussian plume model. The model describes dispersal over distances up to 10 km downwind from a point source. Spore concentrations at a given point depend on the distance from the source, the wind direction, the number of released spores, the wind speed and the amount of mixing in the atmosphere as affected by weather conditions and the effects of the vegetation on the wind flow [59].

Gaussian models may apply to heavy gas dispersion if the wind speed is sufficiently high, and at sufficient distance downstream for the details of the source behavior to be unimportant [60]. Because of its flexibility, a Gaussian dispersion model was adapted for use in the chemical hazard simulation module of SafeCity.

The following analysis depicts a hypothetical scenario of a cloud of toxic fumes rising from a chemical plant near the Vistula river. The simulation is modeled after an actual accident in which some amounts of non-toxic waste were released from the nearby storehouse.

The modeled scenario involves the evaporation of 100 kg of toxins from the storehouse, in the presence of South-Western wind with the speed of 12 km/h. When this information is input through the administrative panel, it is dispatched by the front controller Servlet to the chemical leakage simulation module. In response, the module generates vector layers containing the maximum extent of the toxic plume under the given conditions. Although the module uses the Gaussian plume model for calculating the shape and maximum reach of the toxic cloud, the final shape of the produced plume may be presented in a form which adheres to the NATO ATP-45 norm on Reporting Nuclear Detonations, Biological and Chemical Attacks, and Predicting and Warning of Associated Hazards and Hazard Areas [61], which is employed by Polish rescue services. Fig. 4 shows the results of the performed simulation visualized by both methods.

The two methods of visualization identify the core affected area in the same manner, with certain differences in determining its maximum extent. While the Gaussian method aims to correctly simulate the shape of the cloud, the ATP-45 model tries to represent the total area which may be subject to even a small amount of toxins. Similarly to the blast results, layers containing toxic leakage simulations may also be further processed by the spatial analysis module of the system. The contents of the system's spatial geodatabase allow, among other things, for extraction of detailed addresses of endangered households and institutions in the event of an emergency.

The conducted analysis shows that the spore concentration remains discernible even 4 km downwind from the source. Whether this amount of toxins poses any real threat depends on the type of released substance. This characteristic, in turn, is directly related to the source of release, which typically may be a chemical facility or a tank carrier. Given the annual distribution of wind strength and direction over the investigated area (in case of the city of Gdansk, South-Western winds are prevalent across the year), such simulations may constitute a valid argument when making decisions regarding the allowed localizations of storage warehouses and chemical plants on urban development plans.

6.3. Analysis of the city's vulnerability to a flooding hazard

Access to sources of fresh water has always been one of the primary concerns of settlers. Because of this, many cities around the world, in particular those with several centuries of history, are located near a water body such as a river or lake. Because of unstable weather conditions in a changing climate, these cities are under a constant risk of being flooded. The city of Gdansk lies in the estuary of the Vistula river, near the coast of the Baltic sea. With certain areas of the city located below the sea level, the risk of floods requires constant monitoring of river embankments as well as maintaining up-to-date maps of flood risk zones.

The most commonly used method of creating maps of flood risk zones involves mapping water overflows from identified sources (e.g. rivers or water reservoirs) by computing the liquid's hydrographical flow over a Digital Terrain Model (DTM) of the investigated area. The basic algorithm for such simulations assumes that the water level in the source(s) increases by a given value in meters, which in consequence produces an overflow, resulting in the water leaving its original beds. The final areas affected by the flood are calculated for individual cells of the DTM grid in all directions from the source. In every given direction, the water level is extrapolated to surrounding grid cells until a grid cell is found where the topographic elevation in the DTM is higher than the simulated water level [62].

In the SafeCity hazard simulation module, floodzone mapping requires the remote user to define a given set of modeling criteria. The input parameters for the model encompass a Digital Terrain Model of the city and the layers representing existing water bodies (which are treated as a source of overflowing water), both of which have their default values but may be redefined if necessary. Other parameters must be defined by the end user individually for every performed analysis. These include the amount by which the simulated water level is increased over its standard value and the extent of the area for which the simulation will be executed. This last parameter is controlled by the bounding box of the currently visible area. Changing it allows

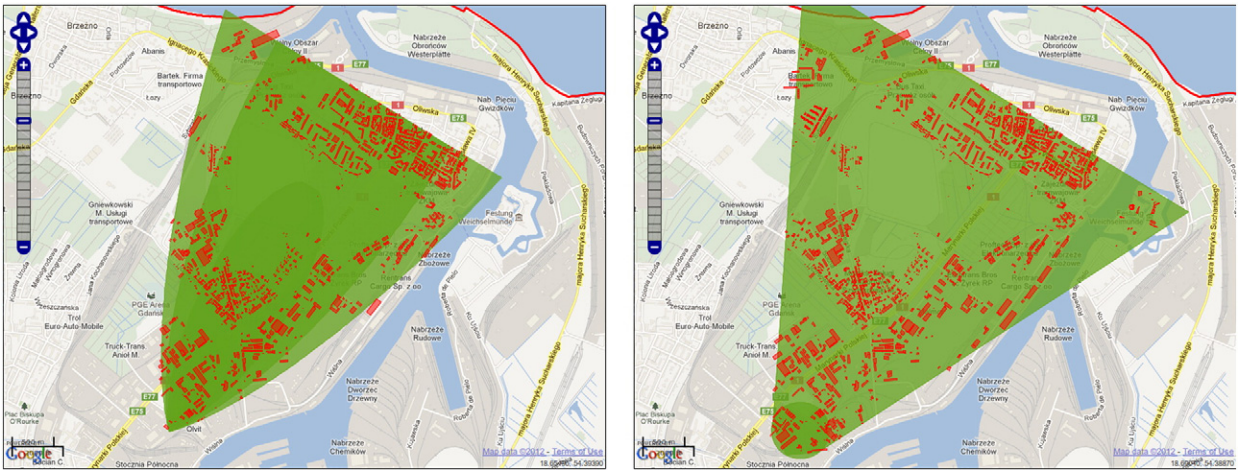

Fig. 4. Evaporation of 100 kg of toxins from a chemical plant in the presence of South-Eastern wind visualized with the Gaussian plume model (left) and the NATO ATP-45 algorithm (right). The chemical puff is represented in green, while the buildings in range of the plume are marked in red.

to optimize the performance of the application, as the computational complexity of the model is heavily dependant on the size of analyzed area.

Fig. 5 shows two results of a simulated increase of the water level in the Vistula river, which crosses the city of Gdansk in several areas, by two meters. The left part of the figure shows results computed for the latest DTM, where recently renovated water dykes (shown in yellow in the background DTM) succeed in stopping the flow of the water under these conditions. In comparison, the right part of the figure presents the outcome of the same simulation executed for DTM data acquired in the year 2003, reconstructing the areas affected by the great flood of 2001, which devastated huge areas of the city.

The main source of flood threat in the city of Gdansk originates from the rise of Vistula water level. It may be caused by a culmination of rainwater from the interior of the country, by strong northern winds pushing the seawater back to the Vistula riverbed, or by both of these at the same time (which was the reason for the tragedy in 2001). The performed analysis shows that, according to detailed DTM data, the recent investments in flood countermeasures ought to prove sufficient to protect the city should the worst-case scenario be reenacted.

6.4. Analysis of the impact of investments in countermeasures on infrastructure criticality

The following analysis involves the application of the system's built-in CARVER2™ vulnerability assessment tool to supporting decisions regarding infrastructure protection. As it has been shown e.g. in Section 6.1 of this article, the spatial proximity of several Critical Infrastructures makes them more susceptible to an attack. The natural solution would be to separate the two institutions geographically. However, this would involve substantial costs and require a considerable amount of time. An alternative would require investing in additional defenses for the existing structures in the hope that it would reduce their vulnerability. In such an event, SafeCity allows one to employ the CARVER2™ module for researching the impact of different countermeasures on the overall resilience of an infrastructure.

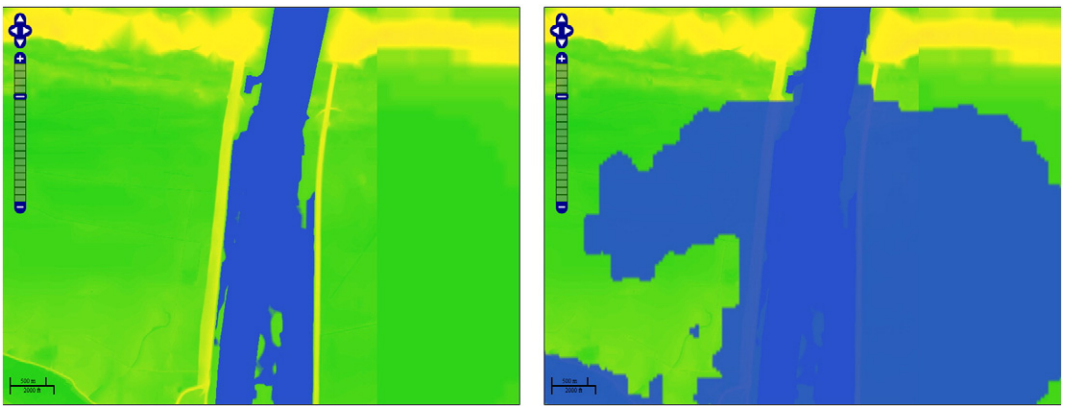

Fig. 5. Flood simulation results before dyke renovation (right) and after dyke renovation (left). Green and yellow colors show the low and high areas in the terrain model respectively. Flooded areas are marked in blue.

The following analysis concerns the situation shown in Fig. 6, where the City Hall Crisis Management Centre (A) is situated in the direct neighborhood to the Thermal power plant (B).

The geographic location of these two infrastructures in close proximity to each other makes both of them prone to a single attack, similar to the one presented in Section 6.1.

The SafeCity CARVER2™ module allows the user to calculate the infrastructure's vulnerability to such an attack. Some basic fields in the CARVER2™ algorithm are automatically filled in with the available information for the selected object. The user may then change and/or manually complete the information in order to perform the calculation. Fig. 7 presents the results of the CARVER2™ algorithm calculated for the Crisis Management Centre.

The total computed blast attack vulnerability score for the Thermal power plant is 387, while the City Hall Crisis Management Centre on its own scores 293 points. Because their proximity makes them prone to a single attack, they may be viewed as a single entity with a combined vulnerability score of 680. The best course of action in this situation would require geographical separation of the two structures. However, such a solution is expensive and time consuming, in particular for infrastructures which need to operate continuously. This being said, in the context of limited resources available for protection of Critical Infrastructures, it may still be possible to enhance the security of these two structures.

The spatial proximity of both infrastructures may be leveraged in a beneficial way by surrounding both of them with a monitored and patrolled perimeter fence. This single investment helps to reduce the criticality score of the City Hall Crisis Management Centre from 293 to 223. Further analysis with the CARVER2™ module reveals that moderate organizational changes related to the Power plant operation by reducing of the number of continually present employees, coupled with the added perimeter fencing, would allow to decrease its vulnerability score to 323. When viewed as a single entity, the total CARVER2™ score of the object drops from 680 to 546 due to the proposed countermeasures.

The above analysis shows that although geographically separating potential targets is the best practice when planning the defense of a city's critical infrastructures, in a situation of limited monetary resources it is possible to significantly reduce the vulnerability of a “cluster” of Critical Infrastructures.

6.5. Analysis of Critical Infrastructure spatial density distribution

As prior analyses have shown, geographic diversification of Critical Infrastructure localizations is an important factor in ensuring proper and continued operation of a city. However, with the increasing number of infrastructures requiring protection, planning their spatial arrangement becomes increasingly difficult. In order to support this process, SafeCity offers a tool for investigating the spread of infrastructures with different levels of criticality over an area, available through its spatial analysis module. The algorithm is derived from standard geographical density estimation, with the addition of an amplitude which corresponds to the criticality of individual infrastructures, as given by their CARVER2™ score. Infrastructures in close proximity to one another are represented by the algorithm in the form of a “cluster” with a criticality level which corresponds to the total value of scores attributed to individual structures.

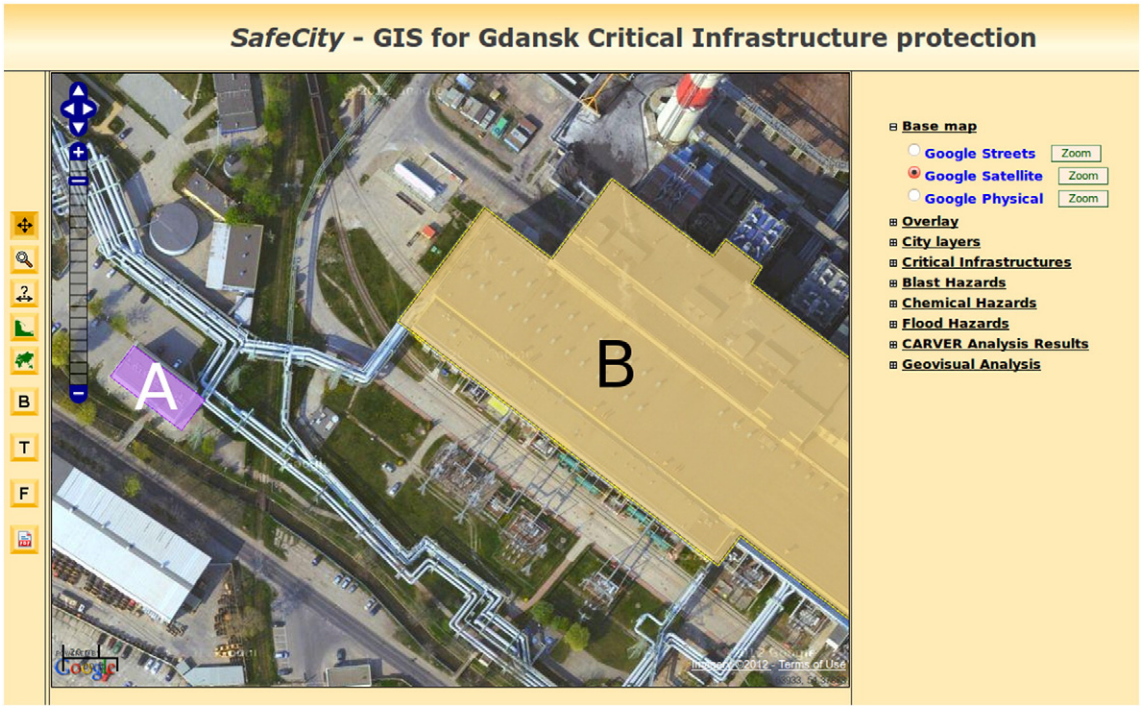

Fig. 6. The City Hall Crisis Management Centre (A) and the Thermal power plant (B) are situated within the same parcel.

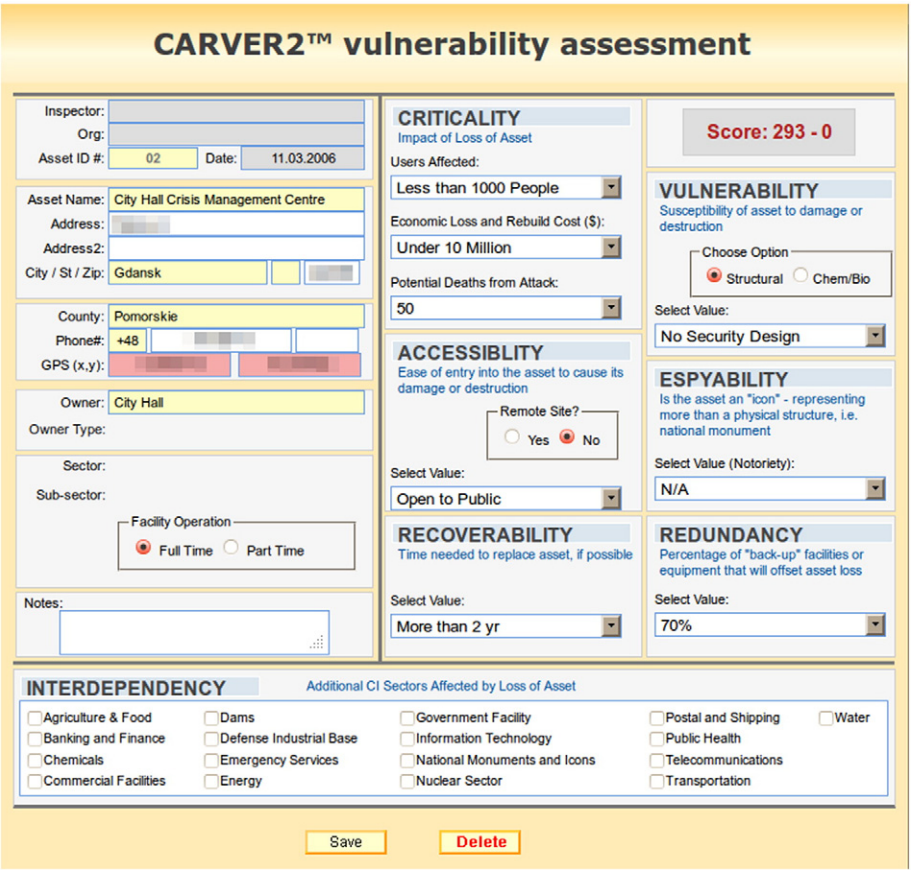

Fig. 7. Window of the CARVER2™ module showing calculations for the Crisis Management Centre.

The following analysis shows the spatial distribution of blast attack vulnerability computed for several infrastructures in the city of Gdansk area.

Fig. 8 presents the spatial distribution of infrastructures with different levels of criticality overlaid on the map of city districts. For the sake of image clarity, the district names have been replaced with unique numbers. As per the paradigms of Geovisual Analytics, the colors of the analysis layer reflect the total vulnerability of local infrastructures. The two red areas in the center of the layer (near district 21) represent highly vulnerable "clusters" of Critical Infrastructure, while the areas marked with orange, yellow and green represent the location and vulnerability of individual structures. The exact correspondence between colors and vulnerability levels will vary for different datasets. In this case red color marks areas where the CARVER2™ score exceeds 700 points, while shades of green denote objects which scored between 100 and 200 points.

The analysis clearly shows a large concentration of highly vulnerable infrastructures in the area between districts 4, 14 and 21. Incidentally, the area also denotes the historical center of the city. Such a situation causes an unnecessary danger to the populace which should be minimized, optimally by moving some infrastructures into other areas. Also according to the analysis, the districts 30, 20 and 12 denote large areas with low criticality and therefore could be designated in land use and urban development plans as good places for locating (or relocating) Critical Infrastructures.

7. Conclusions

Geographical concentration of Critical Infrastructures is an important issue that governments are becoming increasingly aware of. Legislative measures are being taken in order to mitigate some of the negative impact of clustering. The United States Congress, for instance, recognized the need to prevent future concentration of Critical Infrastructure in the aftermath of hurricane Katrina. The first relevant policy it passed was the Energy Policy Act which states that new liquefied natural gas import terminals should be facilitated in diverse ports [63]. However, laws like this do not yet deal with concentration of infrastructures from different sectors. They will not prevent scenarios similar to the recent terrorist attack in Oslo, in which a single determined individual equipped with a car bomb made of fertilizer and fuel oil has succeeded in disrupting the operations of several institutions. While the current legislation in the form of NSHS and EPCIP will not prevent the creation of such clusters, they may be identified and analyzed with the aid of appropriate software.

This research has shown that the application of a dedicated GIS to the analysis of Critical Infrastructure even

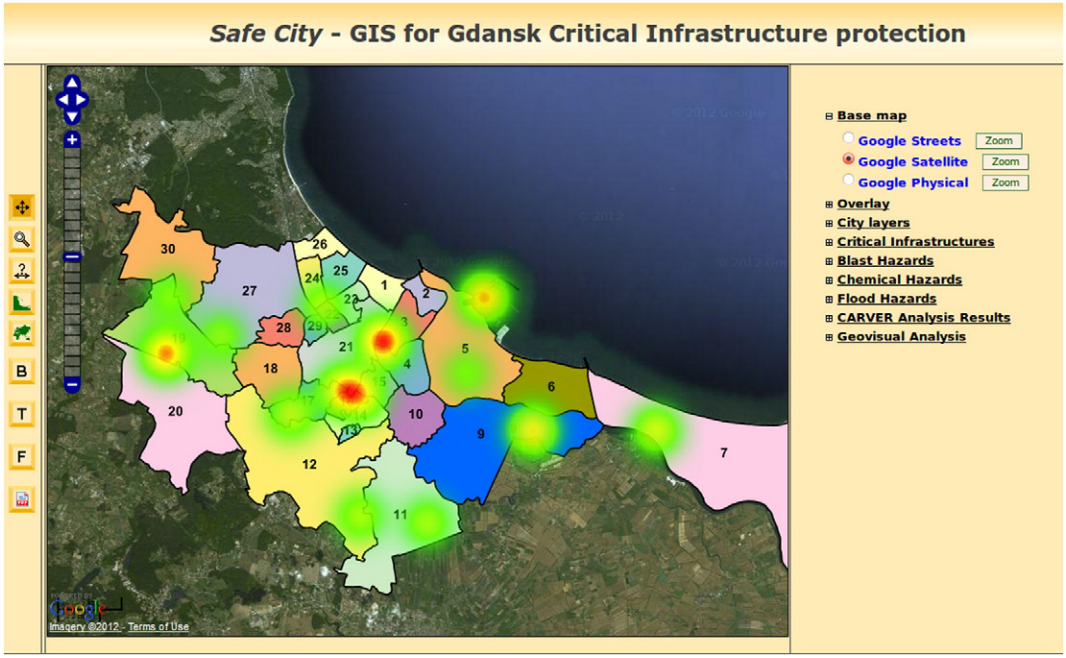

Fig. 8. Analysis of the spatial distribution of Critical Infrastructure vulnerability in Gdansk.

on a very basic level helps to identify often overlooked issues such as the spatial concentration of infrastructures from different sectors. The presented system offers tools which have been designed for specific purposes related to modeling and spatial analysis of different types of hazards in urban regions. The algorithms employed by these tools operate in a spatial context and produce thematic layers representing either the outcome of a possible hazard (in the case of models) or the results of spatial risk analysis. On the basis of the type of produced analysis, each algorithm may be applied to different phases of the emergency management process. In particular, the hazard simulation algorithms may be applied to creation of hazard scenarios during the Preparedness phase, mapping of identified threats in the Response phase as well as simulating the geographical impact of threats during the Mitigation phase.

The integrated CARVER2™ module enables an analyst to identify weaknesses in the defense of Critical Infrastructures by means of target analysis with the use of data available in the city databases. By employing quantitative methods of analysis to a set of standard attributes, the tool allows for comparing dissimilar types of infrastructure using the same standards and aims to assist the user in making decisions regarding investments in countermeasures. The computed criticality scores are stored in the system geodatabase and may be used for further analysis e.g. of the spatial distribution of infrastructure vulnerability. This algorithm helps to identify "clusters" of infrastructures highly susceptible to a specific type of attack. Because such groups of objects may be a much more attractive target than any single infrastructure on its own, their identification should be a priority in strategic planning of resource allocation for security and attack prevention. In the Critical Infrastructure emergency management process, the algorithm is meant to be used during the Mitigation phase.

The SafeCity GIS not only allows for remote execution of diverse analysis algorithms, but also enables rapid dissemination of their results between different parties. For example, an analysis of the possible outcomes of a flood hazard performed by a remote analyst may be readily accessed by rescue services on their way to the affected area. However, in case an official may not be authorized (or qualified) to use the system, the SafeCity GIS also allows for quick creation of an image file containing selected elements (e.g. thematic layers) of the produced analyses. Due to the application of Visual Analytics, the visualizations provided by the developed algorithms convey more information, which should enable quicker decision making. This in turn should increase the amount of time available for taking decisive actions. By providing a central, easy to use hub for storage and analysis of Critical Infrastructure data, the system aims to avoid some of the common bottlenecks blocking the adoption of planning support GIS. In particular, even if public services are aware of the availability of a planning support system, they often either lack the experience to make full use of the system, or simply have no intentions of using it [64]. Because the software was built with the direct involvement of municipalities and rescue services, most of these obstacles should be removed. The sample deployment of the system in the city of Gdansk has shown that applying modern Open Source technology to processing of existing data in cooperation with prospective users results in a cost-effective solution which municipalities are eager to use. This way the system constitutes an invaluable asset in Critical Infrastructure protection, improving synergy between authorities and providing complex support in different phases of the hazard management process.

It should be pointed out that the use of both the presented software and methodology is in no way limited to the

described case of the City of Gdansk, but may be easily applied to any town situated in Poland, in Europe or quite likely any other part of the world which possesses a basic IT infrastructure. This, in particular, applies to small and medium-sized towns which cannot afford an extensive infrastructure security policy. The outcome of the presented study shows that the quality of Critical Infrastructure protection in such cities may be substantially improved at virtually no additional costs.

It should be stressed, however, that although such software offers great help in strategic planning, it will not replace an experienced analyst who understands that eventual investments in countermeasures must be carefully balanced because an antagonist may not necessarily attack the optimal target [65].

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References

[1] R. Hanson, Perspectives on Urban Infrastructure, National Academies Press, 1984.
[2] M. Bulu, O. Ozben, I.H. Eraslan, Clusters in Turkish Textile Industry— A Case Study in Bayrampasa District, II, International Istanbul Textile Congress (IITC), Istanbul, 2004.
[3] J. Madge, The new towns program in Britain, J. Am. Inst. Plann. 28 (1962) 208–219.
[4] S. Graham, Software-sorted geographies, Prog. Hum. Geogr. 29 (2005) 562–580.
[5] J.E. Abbate, From ARPANET to Internet: A History of ARPA-sponsored Computer Networks, 1966–1988, Available at: http://repository.upenn.edu/dissertations/AAI9503730 1994(Accessed May 10, 2013).
[6] S. Graham, Disrupted Cities: When Infrastructure Fails, Routledge, 2009, ISBN 041599179X.
[7] President's Commission on Critical Infrastructure Protection, Critical Foundations: Protecting America's Infrastructures, 1997.
[8] J. Moteff, C. Copeland, J. Fischer, Critical Infrastructures: What Makes an Infrastructure Critical? Congressional Research Service, 2003. Available at: www.fas.org/irp/crs/RL31556.pdf [Accessed June 15, 2012].
[9] R. Radvanovsky, Critical Infrastructure, CRC Press, 2006, ISBN 0849373980.
[10] European Commission, Critical Infrastructure Protection in the Fight against Terrorism, Available at: http://europa.eu/legislation_summaries/justice_freedom_security/fight_against_terrorism/l33259_en.htm(Accessed July 10, 2012).
[11] B. Wisner, P. Blaikie, T. Cannon, I. Davis, At Risk — Natural Hazards, People's Vulnerability and Disasters, Routledge, Wiltshire, 2003, ISBN 0-415-25216-4.
[12] D. Alexander, Principles of Emergency Planning and Management, Terra Publishing, Harpenden, 2002, ISBN 1-903544-10-6.
[13] F.C. Cuny, Disasters and Development, Oxford University Press, Oxford, 1983.
[14] G. Haddow, Introduction to Emergency Management, Butterworth–Heinemann, Amsterdam; Boston, 2003, ISBN 9780750676892.
[15] K. Lin, K.E. Holbert, PRA for vulnerability assessment of power system infrastructure security, Proceedings of the 37th Annual North American Power Symposium, IEEE, 2005, pp. 43–51.
[16] T.E. Palmatier, Building secure critical infrastructures, Int. J. Crit. Infrastruct. Prot. 6 (2013) 85–86.
[17] J. Johansson, H. Hassel, An approach for modelling interdependent infrastructures in the context of vulnerability analysis, Reliab. Eng. Syst. Saf. 95 (2010) 1335–1344.
[18] L. Duenas-Osorio, S.M. Vemuru, Cascading failures in complex infrastructure systems, Struct. Saf. 31 (2009) 157–167.
[19] W.M. Leavitt, J.J. Kiefer, Infrastructure interdependency and the creation of a normal disaster the case of hurricane Katrina and the city of New Orleans, Public Works Manage. Policy 10 (4) (2006) 306–314.
[20] S. Panzieri, R. Setola, Failures propagation in critical interdependent infrastructures, Int. J. Model. Identif. Control. 3 (1) (2008) 69–78.
[21] R. Abdalla, C.V. Tao, Q. Cheng, J. Li, A network-centric modeling approach for infrastructure interdependency, Photogramm. Eng. Remote. Sens. 73 (6) (2007) 681–690.
[22] J.R. Santos, Inoperability input-output modeling of disruptions to interdependent economic systems, Syst. Eng. 9 (1) (2006) 20–34.
[23] H.S.J. Min, W. Beyeler, T. Brown, Y.J. Son, A.T. Jones, Toward modeling and simulation of critical national infrastructure interdependencies, IIE Trans. 39 (1) (2007) 57–71.
[24] T. Brown, W. Beyeler, D. Barton, Assessing infrastructure interdependencies: the challenge of risk analysis for complex adaptive systems, Int. J. Crit. Infrastruct. 1 (1) (2004) 108–117.
[25] G.E. Apostolakis, D.M. Lemon, A screening methodology for the identification and ranking of infrastructure vulnerabilities due to terrorism, Risk Anal. 25 (2) (2005) 361–376.
[26] T. McDaniels, S. Chang, K. Peterson, J. Mikawoz, D. Reed, Empirical framework for characterizing infrastructure failure interdependencies, J. Infrastruct. Syst. 13 (3) (2007) 175–184.
[27] W. Kroger, Critical infrastructures at risk: a need for a new conceptual approach and extended analytical tools, Reliab. Eng. Syst. Saf. 93 (2008) 1781–1787.
[28] J.R. Brock, J. Boltz, E. Doring, M. Gilmore, Information Security Risk Assessment: Practices of Leading Organizations, Available at: U.S. General Accounting Office, 1999. (http://www.gao.gov/products/AIMD-00-33 [Accessed July 12, 2012]).
[29] A. Bobcow, K.T. Kosmowski, Methodological issues of security vulnerability analysis and risk assessment, Konbin 6 (2008) 89–110.
[30] T.G. Lewis, Critical Infrastructure Protection in Homeland Security, John Wiley and Sons, 2006, ISBN 0471786284. (9780471786283).
[31] Center for Chemical Process Safety, Guidelines for Analyzing and Managing the Security Vulnerabilities of Fixed Chemical Sites, Wiley–American Institute of Chemical Engineers (AIChE), 2003, ISBN 081690877X.
[32] PS&S Interagency Working Group, ASME Homeland Security, Risk Analysis and Management for Critical Assets Protection (RAMCAP) Methodology Document, ASME Critical Assets Protection Initiative, 2004.
[33] D.J. Landoll, The Security Risk Assessment Handbook, CRC Press, 2006, ISBN 0849329981. (9780849329982).
[34] P. Pederson, D. Dudenhoeffer, S. Hartley, M. Permann, Critical infrastructure interdependency modeling: a survey of US and international research, Idaho National Laboratory Report INL/EXT-06-11464, 2006.
[35] The Integrated System Model. Electrical Distribution Design, Available at: http://edd-us.com/PubArticles/WebBrochures/ISM_OverviewBrochure.pdf (Accessed July 17, 2013).
[36] TRANSIMS, An Open Source Transportation Modeling and Simulation Toolbox, Available at: http://code.google.com/p/transims/ (Accessed July 16, 2013).
[37] V. Cardellini, E. Casalicchio, E. Galli, Agent-based modeling of interdependencies in critical infrastructures through UML, Proceedings of the 2007 Spring Simulation Multiconference, vol.2, 2007, pp. 119–126.
[38] E. Bagheri, H. Baghi, A.A. Ghorbani, A. Yari, An agent-based service-oriented simulation suite for critical infrastructure behaviour analysis, Int. J. Bus. Process. Integr. Manag. 2 (4) (2007) 312–326.
[39] D.D. Dudenhoeffer, M.R. Permann, M. Manic, CIMS: a framework for infrastructure interdependency modeling and analysis, Proceedings of the 38th Conference on Winter, Simulation, 2006, pp. 478–485.
[40] K.G. Crowther, Y.Y. Haimes, Application of the inoperability input–output model (IIM) for systemic risk assessment and management of interdependent infrastructures, Syst. Eng. 8 (4) (2005) 323–341.
[41] M. Taylor, Identifying Vulnerability, American International Security Corporation, 2002.
[42] B.T. Bennett, Understanding, Assessing, and Responding to Terrorism: Protecting Critical Infrastructure and Personnel, John Wiley & Sons, 2007, ISBN 0-471-77152-X.
[43] R. Peimer, Target Analysis and CARVER2 (Infrastructure Protection, Infrastructure Security), Available at: http://www.ni2cie.org/targetanalysis.php.htm (Accessed June 12, 2012).
[44] National Infrastructure Institute Center for Infrastructure Expertise CARVER2, National Infrastructure Institute, Available at: http://www.ni2cie.org/downloads/Carver2webflyer.pdf (Accessed July 16, 2012).
[45] M. Kulawiak, Algorithms for Processing and Visualization of Critical Infrastructure Security Data as Well as Simulation and Analysis of Threats, Ph.D. dissertation Gdansk University of Technology, 2010.
[46] J. Thomas, K. Cook, Illuminating the Path: The Research and Development Agenda for Visual Analytics, National Visualization and Analytics Ctr, 2005, ISBN 0769523234.
[47] G. Andrienko, N. Andrienko, P. Jankowski, D. Keim, M.J. Kraak, A. MacEachren, S. Wrobel, Geovisual Analytics for spatial decision support: setting the research agenda, Int. J. Geogr. Inf. Sci. 21 (2007) 839–857.

[48] T. Kokkonen, T. Ihaksi, A. Jolma, S. Kuikka, Dynamic mapping of nature values to support prioritization of coastal oil combating, Environ. Model. Softw. 25 (2) (2010) 248–257.
[49] O. Huisman, I.F. Santiago, M.J. Kraak, B. Retsios, Developing a Geovisual Analytics environment for investigating archaeological events: extending the space–time cube, Cartogr. Geogr. Inf. Sci. 36 (3) (2009) 225–236.
[50] B.M. Tomaszewski, A.C. Robinson, C. Weaver, M. Stryker, A.M. MacEachren, Geovisual Analytics and crisis management, Proceedings of the 4th International ISCRAM Conference, Netherlands, 2007, pp. 173–179.
[51] C. Holmes, GeoServer Features. The GeoServer Project, Available at: http://geoserver.org/display/GEOS/Features (Accessed July 22, 2012).
[52] Z. Gui, J. Du, Mapping Distributed Traffic Information Based on GeoRSS&WMS, Proceedings of the 2007 International Conference on Convergence Information Technology, Gyeongbuk, Korea, 2007, pp. 1634–1637.
[53] C. Schmidt, OpenLayers: Free Maps for the Web, http://www.openlayers.org (Accessed July 22, 2012).
[54] M.J. Tang, Q.A. Baker, A new set of blast curves from vapor cloud explosion, Process. Saf. Prog. 18 (1999) 235–240.
[55] R. Turney, R. Pitblado, Risk Assessment in the Process Industries, Institute of Chemical Engineers (IChemE), 1996, ISBN 0852953232.
[56] United States, Dept. of the Army, Fundamentals of Protective Design for Conventional Weapons, Dept. of the Army Headquarters, 1993.
[57] R.K. Zipf, K.L. Cashdollar, Effects of Blast Pressure on Structures and the Human Body, National Institute for Occupational Safety and Health (NIOSH), 2006.
[58] J.V. Ramsdell, G.F. Athey, C.S. Glantz, MESOI Version 2. 0: An Interactive Mesoscale Lagrangian Puff Dispersion Model with Deposition and Decay, NUREG/CR-3344, Pacific Northwest Lab, Richland, WA (USA), 1983.
[59] F. Pasquill, P. Michael, Atmospheric diffusion, 2nd edition, Phys. Today 30 (1977) 55.
[60] J.P. Kunsch, T.K. Fannelop, Unsteady heat-transfer effects on the spreading and dilution of dense cold clouds, J. Hazard. Mater. 43 (1995) 169–193.
[61] NATO ATP-45 Reporting Nuclear Detonations, Biological and Chemical Attacks, and Predicting and Warning of Associated Hazards and Hazard Areas, Available at: http://aero-defense.ihs.com/document/abstract/JMLTJBAAAAAAAAAA (Accessed June 25, 2012).
[62] P. Willems, G. Vaes, D. Popa, L. Timbe, J. Berlamont, Quasi 2D river flood modelling, River Flow 2002 (2) (2002) 1253(ISBN 9058095096).
[63] P.W. Parfomak, Vulnerability of Concentrated Critical Infrastructure: Background and Policy Options, Library of Congress, Congressional Research Service, Washington DC, 2005.
[64] G. Vonk, S. Geertman, P. Schot, Bottlenecks blocking widespread usage of planning support systems, Environ. Plann. A 37 (2005) 909–924.
[65] E. Jenelius, J. Westin, A.J. Holmgren, Critical infrastructure protection under imperfect attacker perception, Int. J. Crit. Infrastruct. Prot. 3 (2010) 16–26.

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